

From Local Curvature to the Weil Functional: An Explicit Construction

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Research Note · March 2026 · v3

Abstract

Building on the curvature decomposition established in [1], we construct an explicit family of admissible test functions $g_{\sigma,\varepsilon}^* \in S_{ad}$ for the Weil explicit formula. For this family we derive a prime-side identity of the form

$$W(g_{\sigma,\varepsilon}^* * \check{g}_{\sigma,\varepsilon}^*) = Z(g_{\sigma,\varepsilon}^*) - H_{\text{local}}(\sigma, \kappa) + O(\varepsilon),$$

where $Z(g_{\sigma,\varepsilon}^*)$ denotes the corresponding zero sum. We further introduce renormalized prime weights $c_p^{\text{ren}} > 0$, prove convergence of the associated series

$$D = \sum_p (c_p^{\text{ren}})^2 \approx 9.471,$$

and show that the renormalized prime-side scale is logarithmic in κ . This isolates the remaining open problem as a single spectral inequality relating $Z(g_{1/2,\varepsilon}^*)$ to $H_{\text{local}}(1/2, \kappa)$. We also record the appearance of the Mertens constant as the natural scale of the renormalized curvature remainder.

1 Introduction and relation to Paper 1

In [1] we studied the curvature field

$$H_\xi(\sigma, t) := \frac{d^2}{d\sigma^2} \log |\xi(\sigma + it)|^2 = H_{\text{local}}(\sigma, \kappa) + H_{\text{dual}}(\sigma, t, \kappa),$$

where $H_{\text{dual}} := H_\xi - H_{\text{local}}$ is the spectral remainder encoding the zero contributions (sign convention as in [1], equation (5)). where the truncated local curvature

$$H_{\text{local}}(\sigma, \kappa) = \psi_1(\sigma) + \sum_{p \leq \kappa} V_p(\sigma)$$

contains the archimedean term and the finite-place contributions. The prime-side pieces satisfy $V_p(\sigma) \geq 0$, and on the critical line one has $H_{\text{local}}(1/2, \kappa) \asymp (\log \kappa)^2$.

The central open question of [1, §7] asks whether this local positivity and critical divergence can be packaged into a global Weil-type positivity statement. The present note addresses a concrete part of that question by constructing an explicit family of admissible test functions

$g_{\sigma,\varepsilon}^* \in S_{ad}$ and identifying the corresponding prime-side term in the Weil functional with the local curvature studied in Paper 1.

The remaining open step is then isolated as a single spectral inequality; see Section 6.

2 Construction of the admissible test function

2.1 The raw family

Recall from [1, Eq. (3)] that

$$V_p(\sigma) = 4(\log p)^2 \frac{p^{-2\sigma}}{(1 - p^{-2\sigma})^2}.$$

Define

$$c_{p,\sigma} := V_p(\sigma)^{1/2} \cdot \frac{p^{1/4}}{(\log p)^{1/2}},$$

and let

$$\varphi_\varepsilon(x) = \frac{1}{\varepsilon\sqrt{2\pi}} e^{-x^2/(2\varepsilon^2)}$$

be the standard Gaussian bump. For fixed κ set

$$\hat{f}_{\sigma,\varepsilon}(x) = \sum_{p \leq \kappa} c_{p,\sigma} \varphi_\varepsilon(x - \log p).$$

The admissible test function is obtained by antisymmetrization:

$$\hat{g}_{\sigma,\varepsilon}^*(x) := \hat{f}_{\sigma,\varepsilon}(x) - \hat{f}_{\sigma,\varepsilon}(-x). \quad (1)$$

Lemma ($g_{\sigma,\varepsilon}^* \in S_{ad}$). *For every $\sigma > 0$ and $\varepsilon > 0$, the function $g_{\sigma,\varepsilon}^*$ belongs to the admissible class S_{ad} used in Lagarias' formulation of Weil positivity.*

Proof. The Gaussian bumps are Schwartz, and antisymmetrization preserves the Schwartz class. Hence $g_{\sigma,\varepsilon}^*$ and $\hat{g}_{\sigma,\varepsilon}^*$ are rapidly decreasing. Moreover, $\hat{g}_{\sigma,\varepsilon}^*(0) = 0$ by antisymmetry, and likewise $g_{\sigma,\varepsilon}^*(0) = 0$. These are the required vanishing conditions at the origin. Finally, the Gaussian decay ensures absolute convergence of the prime-side and zero-side expressions appearing in the Weil functional for this family. $\square$

3 Prime-side identity for the Weil functional

Proposition (Prime-side identity). *For the family $g_{\sigma,\varepsilon}^*$ constructed above, one has*

$$W(g_{\sigma,\varepsilon}^* * \check{g}_{\sigma,\varepsilon}^*) = Z(g_{\sigma,\varepsilon}^*) - H_{\text{local}}(\sigma, \kappa) + O(\varepsilon), \quad (2)$$

where $Z(g_{\sigma,\varepsilon}^*) = \sum_\rho |\hat{g}_{\sigma,\varepsilon}^*(\rho)|^2$ runs over the nontrivial zeros of $\zeta(s)$.

Proof sketch. Apply the Weil explicit formula in the Lagarias framework. The constant term vanishes because $\widehat{g}_{\sigma,\varepsilon}^*(0) = 0$. By construction of the coefficients $c_{p,\sigma}$, the prime-side quadratic form converges to $H_{\text{local}}(\sigma, \kappa)$ as $\varepsilon \rightarrow 0$. The error term $O(\varepsilon)$ arises from the Gaussian smoothing of the prime-power contributions and is not made uniform in κ in the present note. $\square$

Remark. Equation (2) is the key bridge between the curvature field of Paper 1 and the Weil positivity framework. The archimedean remainder is exactly zero since $\widehat{g}_{\sigma,\varepsilon}^*(0) = 0$ (condition S2 of S_{ad}).

4 Renormalized weights and convergence

4.1 Positivity of the renormalized coefficients

At $\sigma = 1/2$ define

$$f_p := V_p(1/2) - \frac{4(\log p)^2}{p}.$$

A direct calculation gives

$$f_p = \frac{4(\log p)^2(2p-1)}{p(p-1)^2} > 0 \quad (p \geq 2). \quad (3)$$

Lemma (Renormalized coefficients are positive). *For every prime $p \geq 2$, the quantity f_p in (3) is strictly positive.*

Proof. The numerator $2p-1 > 0$ and the denominator $p(p-1)^2 > 0$ for every prime $p \geq 2$. $\square$

The renormalized coefficients are

$$c_p^{\text{ren}} := f_p^{1/2} \frac{p^{1/4}}{(\log p)^{1/2}} > 0. \quad (4)$$

Numerical values for small primes:

p	$V_p(1/2)$	$4(\log p)^2/p$	f_p	$[c_p^{\text{ren}}]^2$
2	5.545	0.962	4.583	2.883
3	3.625	1.600	2.025	2.012
5	3.213	2.058	1.155	1.166
7	3.019	2.301	0.718	0.781
11	2.830	2.570	0.260	0.439

4.2 Convergence of the renormalized energy

Lemma (Convergence). *The series*

$$D := \sum_p [c_p^{\text{ren}}]^2 = \sum_p \frac{4(\log p)^2(2p-1)}{p(p-1)^2} \quad (5)$$

converges absolutely. Numerically $D \approx 9.471$ (primes below 10^4 ; tail < 0.009).

Proof. For large p , $[c_p^{\text{ren}}]^2 \sim 8(\log p)^2/p^2$. Since $\sum_n (\log n)^2/n^2 < \infty$, convergence follows by comparison. $\square$

5 The renormalized scale and the Mertens constant

By the prime number theorem,

$$H_{\text{local}}(1/2, \kappa) = 2(\log \kappa)^2 + A\gamma_M \log \kappa + C_0 + o(\log \kappa),$$

where $\gamma_M \approx 0.2615$ is the Mertens constant,

$$\gamma_M = \gamma + \sum_p \left[\log \frac{p}{p-1} - \frac{1}{p} \right],$$

and $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. The renormalized remainder

$$H_{\text{local}}^{\text{ren}}(1/2, \kappa) := H_{\text{local}}(1/2, \kappa) - 2(\log \kappa)^2 \sim A\gamma_M \log \kappa$$

lives on the logarithmic scale, and the Mertens constant is its natural coefficient.

Remark. The Euler–Mascheroni constant γ appears simultaneously in γ_M (prime side, via the curvature remainder) and in the first Li coefficient $\lambda_1 = 1 + \gamma/2 - \log 2 - (\log \pi)/2 \approx 0.0231$ (zero side). This structural coincidence reflects the universal role of γ as the renormalization constant governing the transition from discrete prime sums to continuous logarithmic integrals.

Remark. The renormalized construction clarifies the scale of the prime-side contribution, but does not by itself resolve the spectral inequality (6).

6 The remaining spectral inequality

For the family $g_{1/2, \varepsilon}^*$, Weil positivity reduces to

$$Z(g_{1/2, \varepsilon}^*) \geq H_{\text{local}}(1/2, \kappa). \tag{6}$$

Open question. Is there an unconditional lower bound

$$Z(g_{1/2, \kappa}^*) = \sum_{\rho} |\widehat{g}_{1/2, \kappa}^*(\rho)|^2 \geq C \cdot (\log \kappa)^2$$

for some constant $C > 0$, where the sum runs over nontrivial zeros of ζ ?

Formally, treating the zeros as independent via Selberg’s orthogonality principle [8] yields

$Z(g^*) \sim D \log \kappa \log \log \kappa$ with $D \approx 9.471$. More precisely: the diagonal contribution

$$\sum_p [c_p^{\text{ren}}]^2 \cdot I(\varepsilon) \sim D \cdot \log \kappa$$

is unconditional (it follows from the prime number theorem alone); the off-diagonal control of the cross-terms $\sum_\gamma \cos(\gamma \log(p/q))$ for $p \neq q$ remains open. An unconditional proof of even $Z \geq C(\log \kappa)^2$ is presently unavailable.

Remark (The constant $D/2\pi$). *The asymptotic $Z(g^*) \sim D \log \kappa \log \log \kappa$ yields a natural bridge constant*

$$\frac{D}{2\pi} \approx \frac{9.471}{6.283} \approx 1.507,$$

which measures the ratio of the prime-side energy D to the archimedean normalization 2π of the zero density $N(T) \sim T \log T/(2\pi)$. This constant is of a different nature from the pair-correlation statistics studied in the GUE conjecture (Montgomery 1973): it depends only on the total density of zeros via the classical formula for $N(T)$, which is unconditional and requires neither GUE nor RH as input. Whether $D/2\pi$ admits a representation in terms of classical constants is an open question.

Remark. *By Lagarias [3], $\text{RH} \iff W(f * \check{f}) \geq 0$ for all $f \in S_{ad}$. A positive answer to the open question above would establish $W(g_{1/2,\kappa}^* * \check{g}_{1/2,\kappa}^*) \geq 0$ for the specific family constructed here, which is a necessary but not sufficient condition for RH in the absence of a density argument for this family in S_{ad} .*

Computational note

All numerical claims are reproducible. Code is available at [2]. The Gaussian parameter $\varepsilon = 0.05$ is required for meaningful numerical experiments with the zero sum (for $\varepsilon = 0.5$ one has $e^{-\varepsilon^2 \gamma_1^2} \approx 10^{-22}$, effectively suppressing all zero contributions).

References

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Acknowledgment

The author thanks Jean-François Burnol for confirming that the curvature decomposition of [1] does not appear in the literature known to him.